

PROJECT PRESENTATION · 2026

AI-Based **Real-Time Object Detection** and Perception System for Autonomous Vehicles

AI

+

OBJECT DETECTION

+

ENVIRONMENTAL PERCEPTION

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01 · INTRODUCTION

What is Autonomous-Vehicle Perception?

Autonomous vehicles must continuously understand their surroundings to operate safely. A **perception system** processes information from sensors such as cameras and converts it into meaningful information about the environment.

OUR PROPOSED SYSTEM AIMS TO

- ✓ Analyze road scenes using visual input
- ✓ Detect important objects around the vehicle
- ✓ Track objects across consecutive frames
- ✓ Understand position & movement of surroundings
- ✓ Identify objects relevant to the vehicle's path
- ✓ Support safety-related decision making



Key idea — the goal is not just to **detect objects**, but to **understand the surrounding driving environment**.



DETECTION → TRACKING → PERCEPTION → RISK

02 · PROBLEM STATEMENT

Raw images are **not enough**

A camera provides a continuous stream of images — but raw pixels do not tell an autonomous vehicle what actions may be required. **Simply detecting an object is not enough.** The system may need to determine:

❓ **What** object is present?

📍 **Where** is it located?

↔ **How** is it moving?

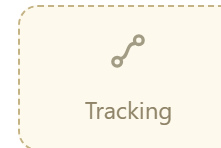
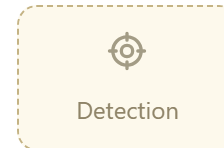
📏 **How close** is it to the vehicle?

🚦 Is it in the **driving region**?

⚠ **Could it create collision risk?**

👁 Object detection, tracking, spatial understanding and safety analysis are often **treated as separate problems.**

⚠ EXISTING CHALLENGE — STUDIED SEPARATELY



Single Real-Time Perception Pipeline

✓ PROPOSED DIRECTION

03 · OBJECTIVES

Six project objectives



01

Real-Time Object Detection

Explore AI-based models such as the **YOLO family** to detect objects in road scenes.



02

Object Tracking

Track detected objects across consecutive frames and maintain their identities.



03

Environmental Perception

Analyze the location and relevance of detected objects within the driving environment.



04

Distance & Movement Analysis

Investigate proximity estimation and analyze object movement over time.



05

Collision-Risk Analysis

Explore metrics such as **Time-to-Collision (TTC)** to identify dangerous situations.



06

Real-Time Visualization

Display detected objects and perception info from recorded video or a live camera.

Five papers shaped the foundation

5 PAPERS REVIEWED**100,000** IMAGES IN BDD100K**10** PERCEPTION TASKS COVERED**BDD100K: A Diverse Driving Dataset for Heterogeneous Multitask Learning**

CVPR 2020 · F. Yu et al.

DATASET

Diverse driving data & multiple perception tasks for training and evaluation

You Only Look Once: Unified, Real-Time Object Detection

CVPR 2016 · J. Redmon et al.

DETECTION

Foundation for fast AI-based object detection

ByteTrack: Multi-Object Tracking by Associating Every Detection Box

ECCV 2022 · Y. Zhang et al.

TRACKING

Maintains object identities across video frames

3D Object Detection for Autonomous Driving: A Comprehensive Survey

IJCV 2023 · J. Mao et al.

3D PERCEPTION

Spatial understanding & different sensing approaches

3D Object Detection for Autonomous Driving: A Survey

Pattern Recognition 2022 · R. Qian et al.

3D DETECTION

Overview of perception methods, datasets and challenges

Literature insight — perception is broader than detection:

DETECTION



TRACKING



SPATIAL



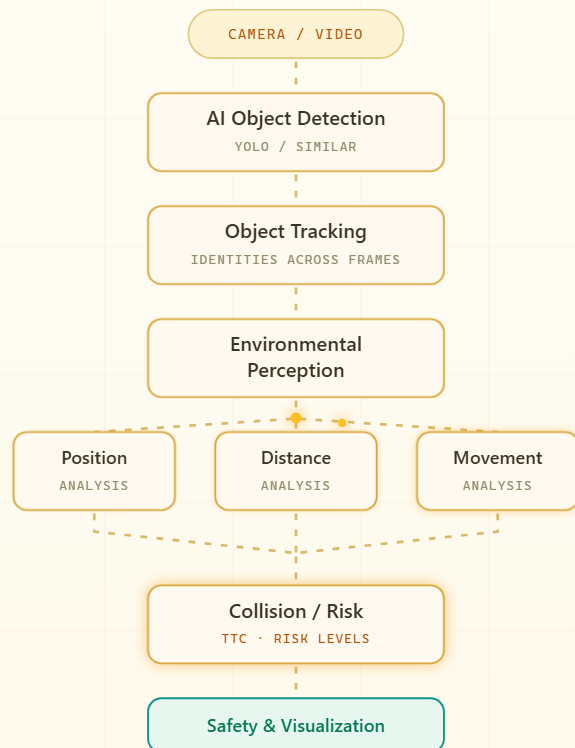
TEMPORAL



SAFETY

05 · PROPOSED SYSTEM

System architecture



- **Input**

Recorded driving video **or** live camera feed, processed frame by frame.

- **Output**

Detected & tracked objects + environmental information + potential safety / risk information.

- 🔗 **Parallel analysis**

Position, distance and movement are analyzed as parallel branches of the same perception layer.

- ⚙️ **Modular design**

Each component can be improved or replaced independently — model, tracker, estimators.

- 🕒 **Real-time path**

The complete pipeline is designed for low-latency, frame-by-frame operation.

06 · METHODOLOGY

How the system will work

1

STEP Capture

Obtain frames from a camera or recorded driving video.

2

STEP Detect

An AI object-detection model identifies road objects and their locations.

3

STEP Track

A multi-object tracker associates objects across consecutive frames.

4

STEP Analyze

Object position · movement · driving-region relevance · approximate distance.

5

STEP Temporal analysis

Information across frames studies motion and can estimate Time-to-Collision (TTC).

6

STEP Risk assessment

Available information is combined to classify the level of collision risk.

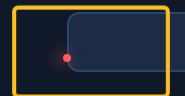
7

STEP Visualization

A real-time interface displays the perceived environment and safety information.

FRAME-BY-FRAME

CAR 0.94 #12



PERSON
0.88
#07



NEAREST OBJECT

12.5 m

CLOSING SPEED

9.3 m/s

TIME-TO-COLLISION

1.4 s

OBJECTS TRACKED

6



▲ WARNING – MONITORING

Risk levels: SAFE · WARNING · DANGER · CRITICAL

PROTOTYPE SAFETY OUTPUT – NOT A CERTIFIED CONTROL SYSTEM

07 · TECHNOLOGY STACK

Proposed technologies & tools



PROGRAMMING
Python



COMPUTER VISION
OpenCV



OBJECT DETECTION
YOLO-family model



OBJECT TRACKING
ByteTrack / similar



DEEP LEARNING
PyTorch · Ultralytics



DATASET
BDD100K / driving dataset

- ✓ **Python** — flexible ecosystem for AI and computer vision.
- ✓ **OpenCV** — camera, video processing and visualization.
- ✓ **YOLO-family** — suited to real-time object detection.
- ✓ **Trackers** — follow objects across frames; **driving datasets** give realistic road scenes.

⚠ **Note** — final model and technology choices will be determined during development and experimentation.



Why YOLO? Speed matters

Reported inference speed — frames per second

Two-stage detectors ≈ 5 FPS



YOLO (2016) 45 FPS



Fast YOLO 155 FPS



As reported in the original YOLO paper (CVPR 2016).

Real-time throughput is a core requirement: a vehicle cannot wait seconds to understand what is in front of it.

What we expect to **achieve** — and where it can go

✓ Expected outcomes

A working prototype capable of:

- ✓ Detecting relevant road objects in real time
- ✓ Tracking objects across consecutive frames
- ✓ Identifying objects in the vehicle's driving region
- ✓ Providing approximate spatial / proximity information
- ✓ Analyzing object movement over time
- ✓ Exploring collision-risk & TTC-based analysis
- ✓ Supporting recorded video and live-camera input
- ✓ Presenting perception through a visual interface

📍 Future scope

The system could later be extended with:



Lane detection



Traffic-light & sign recognition



Advanced depth estimation



LiDAR & radar integration



Multi-sensor fusion



Vehicle trajectory prediction



Pedestrian movement prediction



Advanced collision prediction



Steering & braking integration

09 · CONCLUSION

From detection to **understanding**

Autonomous vehicles need more than simple object recognition — they must understand **objects, positions, movement, and their relationship with the vehicle's path**. Our project begins with visual object detection and progressively grows into a broader **real-time perception system**.

AI OBJECT DETECTION



OBJECT TRACKING



ENVIRONMENTAL PERCEPTION



DISTANCE & MOVEMENT



COLLISION-RISK ASSESSMENT

Questions?

THANK YOU ·

AI-BASED PERCEPTION FOR AUTONOMOUS VEHICLES